



Address: Dariaus ir Girėno g. 32A-1, LT-02188 Vilnius
e-mail: info@lifelab1.com
License No.: 4864

Protocol generation date:
No.:

LABORATORY TEST PROTOCOL

Patient code:
Age:
Sex:

Sample type: dried blood spot (DBS)
Sample collected:
Sample received:

Test	Analysis date	Result	Measurement units	Interpretation	Method	Employee license No.
Haemoglobin A1c (HbA1c)			mmol/mol		1	
			%			

Analysis method:

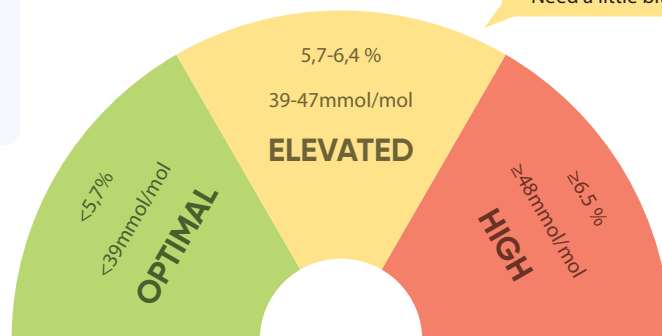
1 – Ion-exchange high-performance liquid chromatography (HPLC)

Hello! Just got your HbA1c test results?

Exciting, right? Let's take a closer look and interpret the HbA1c test together!

Your HbA1c result reflects your average blood glucose levels over the past 3 months. Now, let's identify which category your HbA1c level falls into and explore the associated recommendations

HbA1c is expressed in National Glycohemoglobin Standardization Program (NGSP) units – %, and International Federation of Clinical Chemistry and Laboratory Medicine (IFCC) units – mmol/mol.



You're doing really great!

Need a little bit of help?

It's time to make changes

Recommendations:



The risk of developing type 2 diabetes is low. In the absence of risk factors, repeat the test after 1 year.



Increased risk of developing type 2 diabetes. Contact your doctor for more detailed examination, treatment and / or lifestyle correction. Repeat the test after 3 months.



Immediately consult a doctor. Repeat the test every 3 months.

How can knowing your HbA1c level benefit you?

By understanding your HbA1c level you get a comprehensive picture of your risk for developing type 2 diabetes, a condition that affects millions worldwide.

Factors such as:

- diet;
 - the amount and type of physical activity;
 - underlying medical conditions;
- can all influence your HbA1c levels.

HbA1c awareness: your defense against diabetes complications

Here's the silver lining: catching a rising HbA1c level provides a great opportunity to prompt a lifestyle reassessment. Taking early action can prevent or slow diabetes and its complications.

Moreover, as you embark on healthier habits, periodic HbA1c measurements become your compass. They show you the tangible effects of dietary changes, increased physical activity, or prescribed medications. It's empowering to see those numbers improve, proving that the efforts you put in towards a healthier life are indeed paying off.

So, your HbA1c levels are on the rise?

No Worries! We've got a friendly guide to help you turn things around



Did you know that getting a good night's sleep is like a superhero move against the risk of type 2 diabetes?

It's not just about the hours: uninterrupted, full sleep cycles are key! So, aim for a cozy, uninterrupted sleep – your body will thank you, and it's a fun way to keep diabetes at bay! [1,2]



Meet aerobic exercise – the friendly wizard for your body!

When you get moving, it not only burns stored fat but also does wonders for your blood glucose levels. Picture this: your muscles working their charm, improving insulin sensitivity, and making your pancreas work its best. All of this teamwork can help bring down your HbA1c levels! Try out cool exercises like running, swimming, or even dancing.

Bonus: Exercise is a stress reliever too! [3]

So, let's wave our magical exercise wands and keep those sugar levels in check!



Meet another champion in our story – nutrition.

It plays a crucial role in well-being, improving outcomes, and bringing down HbA1c levels. In a study, folks who rocked the recommended levels of fruits and veggies had a great association with HbA1c control! [4] And You know what? Dietary fiber acts like a fighter slowing down glucose absorption after those tasty, high-carb meals, giving your body a break. Less insulin demand, more superpower vibes! [5] So, let's keep those plates colorful with fruits and veggies – your body will high-five you for it!



Also, ever checked your vitamin D levels?

Turns out, there's a significant connection with HbA1c. Scientists have presented compelling evidence highlighting a noteworthy inverse association between vitamin D and HbA1c. So, if your HbA1c is on the higher side, it might be worth checking your vitamin D levels. [6] Correcting any deficiency can be a game-changer for blood glucose control and your overall health.

Literature

1. Kass L, Sanderson JC, Desai T & Hurst R (2022) The relationship between the elevation of haemoglobin A1c level, sleep quality and sleep duration in clinically diagnosed pre-diabetic patients in a nationally representative sample. *Diabetes Vasc Dis Res* **19**, 14791641211067420.
2. Keskin A, Ünalacak M, Bilge U, Yildiz P, Güler S, Selçuk EB & Bilgin M (2015) Effects of Sleep Disorders on Hemoglobin A1c Levels in Type 2 Diabetic Patients. *Chin Med J (Engl)* **128**, 3292–3297.
3. Singh B, Koneru YC, Zimmerman H, Kanagala SG, Milne I, Sethi A & Jain R (2023) A step in the right direction: exploring the effects of aerobic exercise on HbA1c reduction. *Egypt J Intern Med* **35**, 58.
4. Elfaki FA, Chandika RM, Kahlani SH, Hakami HH, Hakami AS, Alsayegh AA, Dighriri AY & Khalafalla HE (2023) Dietary patterns and their associations with glycemic control among type 2 diabetic patients in Jazan, Saudi Arabia: A cross-sectional study. *Medicine (Baltimore)* **102**, e34296.
5. Goff HD, Repin N, Fabek H, El Khoury D & Gidley MJ (2018) Dietary fibre for glycaemia control: Towards a mechanistic understanding. *Bioact Carbohydrates Diet Fibre* **14**, 39–53.
6. Buhary BM, Almohareb O, Aljohani N, Alrajhi S, Elkaissi S, Sherbeeni S, Almaghamisi A, Khan SA & Almalki MH (2017) Association of Glycosylated Hemoglobin Levels With Vitamin D Status. *J Clin Med Res* **9**, 1013–1018.